# Mechanisms of twinning: VII. Effect of diet and heredity on the human twinning rate.

OBJECTIVE: To evaluate the possible biochemical effect of diet and heredity on the rates of monozygotic and dizygotic twinning. STUDY DESIGN: In that insulin-like growth factor (IGF) has been found to be elevated in cows selected for their demonstrated increased twinning rate, the effect of agents that influence the level of IGF in women was examined. This was correlated with their prior history of singleton versus twin birthing. In particular, the effect of diets consisting of or excluding animal products that have elevated IGF content (e.g., milk) was considered. RESULTS: Vegan women, who exclude dairy products from their diets, have a twinning rate which is one-fifth that of vegetarians and omnivores. CONCLUSION: The results reported here support the proposed IGF model of dizygotic twinning. Genotypes favoring elevated IGF and diets including dairy products, especially in areas where growth hormone is given to cattle, appear to enhance the chances of multiple pregnancies due to ovarian stimulation.